Exam Review – MPM 2D

Part 1: <u>Equations and Linear Systems</u>.

1. The line $y = -\frac{5}{8}x + 7$ has a slope ________ and a y-intercept _________.

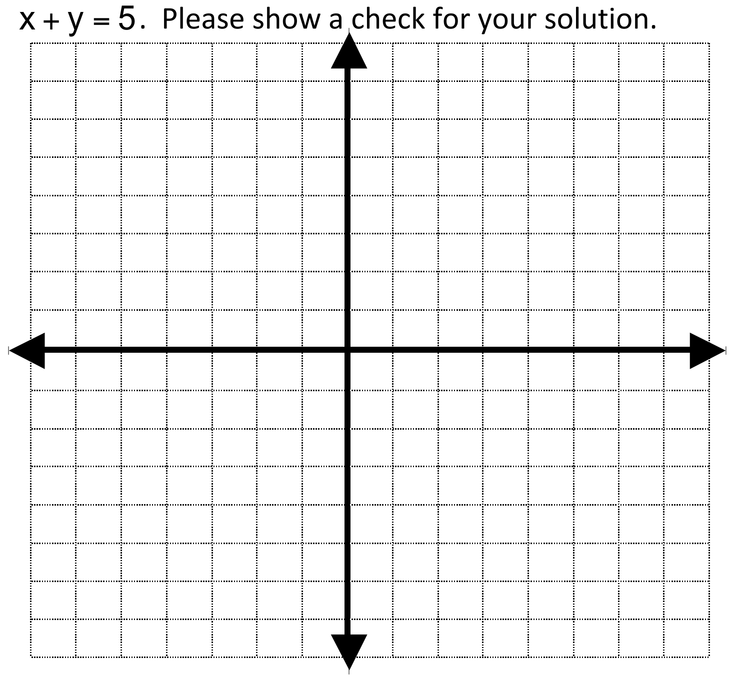
2. Create a linear system that has two equations for lines that do not intersect. Please use $y = mx + b$ form.

3. Is the point $(-4,\ 5)$ the solution to the system $y = -3x - 7$ and $2x - 3y = -23$?

   Explain.

4. Solve the system $\begin{aligned} 2x + 5y &= 7 \\ x + 8y &= -2 \end{aligned}$ by **either substitution or elimination**.

5. **Solve by graphing:** $2x - 3y = 15$ and $x + y = 5$. Please show a check for your solution.



6. How many solutions does the system $\begin{array}{l} y = 2x - 5 \\ 4x + 2y + 20 = 0 \end{array}$ have? (none, infinite, or one)

Explain.

7. Define variables and set up the linear system. Do not solve the problem.

a) **A banquet hall serves two different dinners: a chicken dish that costs $16 and a beef dish that costs $18. The 300 wedding guests ordered their meals in advance and the total cost to prepare the dinners is $5256. How many of each type of dinner is prepared?**

Let x represent ____

Let y represent ____

Equations to represent the problem are ____________________ and ____________________

b) **Iona Bank has $23.50 in loonies and quarters. There are 40 coins in all. How many of each coin are there?**

Let x represent ____

Let y represent ____

Equations to represent the problem are ____________________ and ____________________

8. Solve the following linear systems. Make sure to define your variables:

a) Looney and Ducky raided their piggy banks and together they came up with $55. If all the money was in loonies and toonies and they had 42 coins in all, how many of each type of coin did they find?

b) A tennis club charges an initiation fee and a monthly fee. At the end of one month, a member has paid a total of $260. At the end of six months, she has paid a total of $435. What are the initiation fee and the monthly fee?

Part 2: Analytic Geometry.

1. The circle $x^2 + y^2 = 4$ has centre _________ and radius ___________.

2. The two formulas needed for finding a perpendicular bisector of a line segment are ____________________ (slope / distance / midpoint) to be sure that the line is **perpendicular**

   and ____________________ (slope / distance / midpoint) to be sure that the line is a **bisector**.

3. Parallel lines have slopes that are

   ____

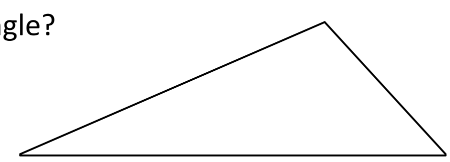
4. Perpendicular lines have slopes that are

   ____

5. For the points $A(9, -10)$ and $B(-3, -2)$ the value for the:
   a) slope AB is:
   b) midpoint of AB is:
   c) length AB is:

   d) slope of a line perpendicular to AB is:
   e) equation of the line through AB is:

6. What is the difference between a median and an altitude in a triangle? Illustrate using the triangle shown.



7. Given any triangle with vertices A, B, and C, explain in words how you would prove that the triangle is a right angled isosceles triangle. Be sure to identify which formulas would help.

8. Triangle ABC has vertices $A(-5, 4)$, $B(3, -2)$, and $C(1, -4)$.

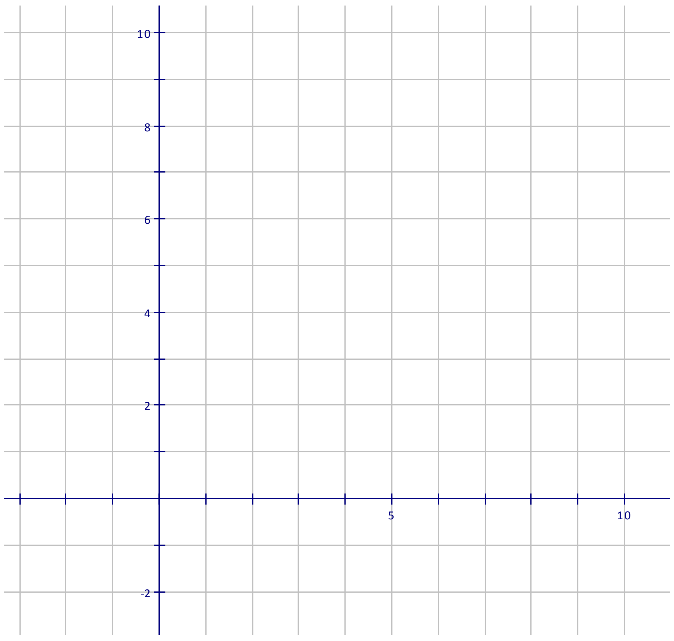
a) Show that $\angle ABC$ is isosceles using an algebraic method.

b) Write the equation for the median from A to BC using an algebraic method.

9. Determine the distance between the point (5, 2) and the line y = 3x -2

Part 3: Quadratics.

1. Graph the equation $y = -\frac{1}{2}(x-3)^2 + 8$ below and answer the following:



a) zeros are ________________

b) vertex is ________________

c) axis of symmetry is ______________

d) max/min value is ___________________

e) the equation in zeros form is

____

2. State the vertex and graph each of the following using as many points that fit on the grid.

a) $y = x^2$

vertex _________

opens:

b) $y = -\frac{1}{2}x^2 - 7$

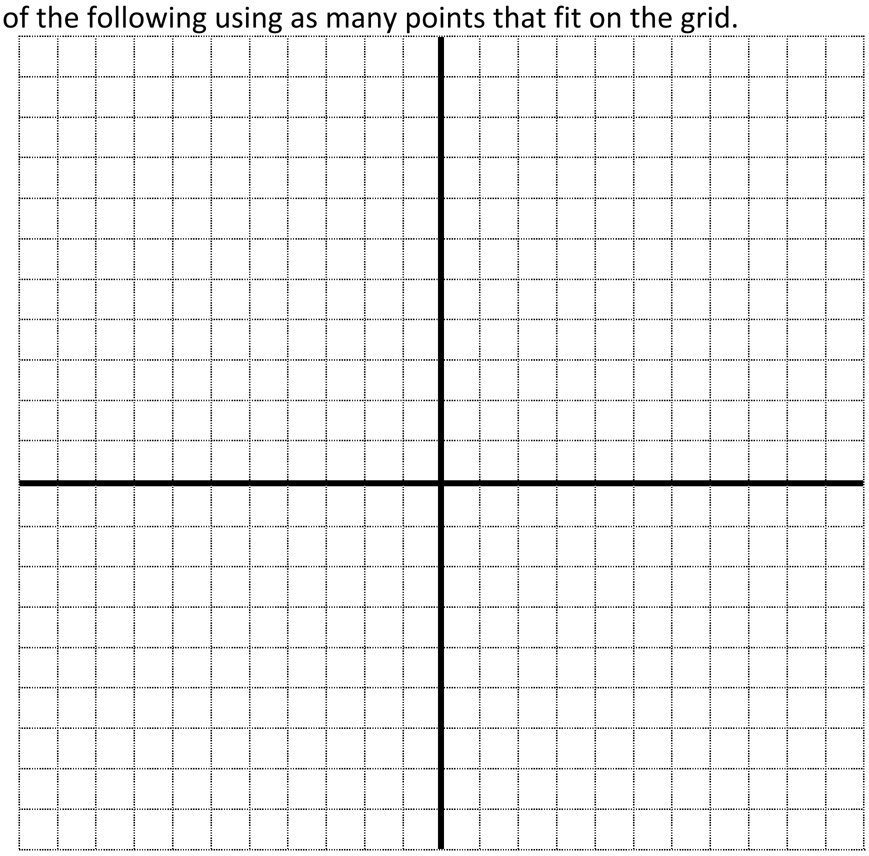
vertex _________

opens:

c) $y = 2(x-7)^2 - 4$

vertex _________

opens:



3. List the transformations in the correct order that makes $y = x^2$ into $y = \frac{1}{2}(x+4)^2 - 3$.

4. Factor: a) $4x^2 - 10x =$ ____________________ b) $x^2 - 16 =$ __________________

c) $x^2 + 9x + 20 =$ __________________ d) $9x^2 - 30x + 25 =$ _________________

e) $3x^2 + 14x - 5 =$ _______________ f) $7x^2 + 35x + 42 =$ ______________

5. Expand (multiply) and simplify (collect)

a) $2(x+5)(x-4)+(x-3)^2$ b) $(x-3)(x-9)$

6. Evaluate. (no decimals)

$$7^0 + 3^{-1} - \left(\frac{1}{3}\right)^2$$

7. For the parabola $y = x^2 - 8x - 9$,
   a) find the zeros, axis of symmetry, and the vertex.
   b) Is the vertex a maximum or a minimum? Explain.

8. Change the quadratic relation to vertex form by completing the square.

$$y = 5x^2 - 30x + 18$$

9. At the end of the course, a student launches his math textbook into the air and its height, h in metres after time, t in seconds can be modelled by the equation $h = -t^2 + 8t + 9$.

a) From what height is the book launched?

b) Factor and find the time when the textbook lands on the ground.

c) What is the maximum height reached by the textbook and when does the book reach this height?

10. The underside of a concrete bridge forms a parabolic arch that is 18 m wide at the water level and 10 m high at the centre.

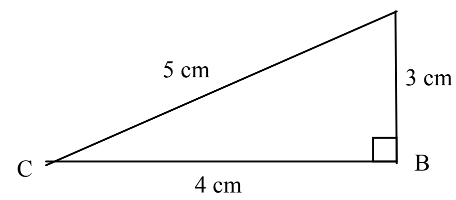
a) Write an equation to model the parabola.

b) Will a sailboat with a mast 8 m tall be able to safely pass if it is 3 m right or left of centre? Use your equation to explain.

Part 4: <u>Similarity and Trigonometry</u>.

A: Fill in the Blanks. [6 marks]

1. For the triangle shown with reference angle C, write the three ratios:



sin C = cos C = tan C =

2. If $\tan A = \frac{7}{25}$, then the measure of angle A to the nearest degree is $\angle A =$______

3. Calculate the measure in the equation to one decimal place. Be sure your calculator is in degree mode.

a) $\sin 65^{o} = \frac{x}{13}$

x = ____________

b) $\cos 31^{o} = \frac{15}{x}$

x = _____________

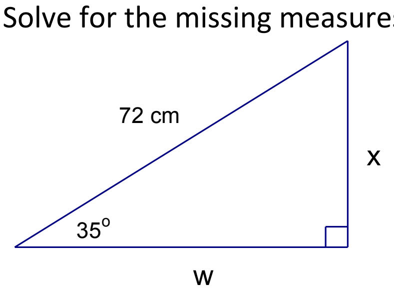
4. Can you use the primary trig functions to solve all types of triangles? Explain and justify your reasoning.

5. To find the height of a tree, Larry places a mirror, face up on the ground, 20 metres from the tree. He then backs away from the mirror and the tree until he can see the top of the tree reflected in the mirror. Larry is 1.2 meters tall and is 0.9 metres from the mirror. How tall is the tree? (to one decimal place). Diagram and proof of similar triangles is required.

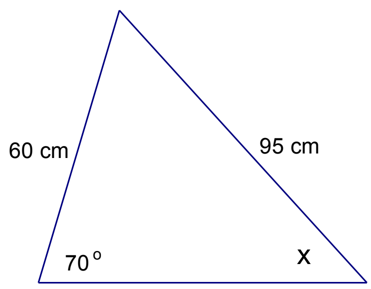
6. An extension ladder is used to paint the trim around the windows on the second and third floors of a building. The ladder is only considered safe if it makes an angle with the ground between 70° and 80°. If a ladder is positioned 3.5 m away from the base of the wall and reaching 8.4 m up the wall, is the ladder being used safely?

7. Solve for the missing measures in each of the triangles. Be sure to explain your work.

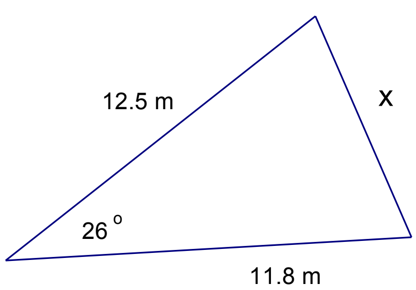
a)



b)

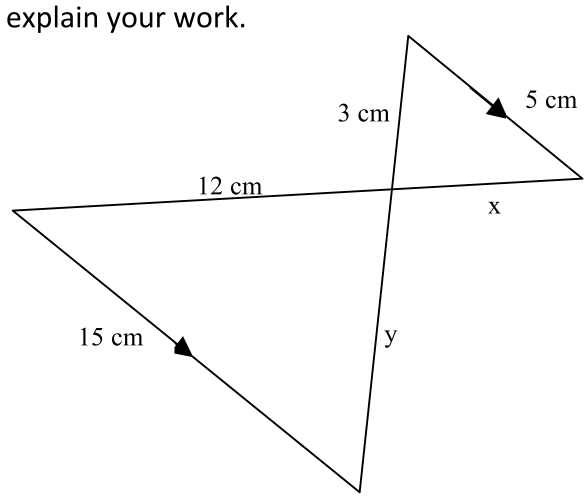


c)



d) ΔABC has $\angle B = 50^{\circ}$, $\angle C = 60^{\circ}$ and $BC = 55$ cm. Solve for the other angles and sides.

8. Solve for the missing measures. Be sure to explain your work.



9. To measure the distance across a pond, a person measures the distance from each end of the pond to a fixed point A as shown. If the angle between these measures is 55°, calculate the distance across the pond to the nearest metre.